An 81-variable quartic under rational *abc*

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Abstract

Under rational *abc* and the specified elliptic-group, prime-selection and pole-realization premises, we construct a family of quartics in 81 total rational variables whose input projection is the integers for every sufficiently large fixed odd coefficient shift. The family is the sum of 73 primitive quadratic squares, has exactly 941 coordinate monomials, and has coefficient height $\max(4153456, B^2 + 797551)$ for every fixed odd $B \geq 3$. The sufficient defining shift is ineffective. The reduction from 87 variables uses two reciprocal elliptic points, recovers their difference instead of storing a separate root point, and permits half-integral indices. A proved trace-height theorem covers these additional points. Additive comparison arguments remove one more stored scale. Identity, tangent and torsion cases are retained until the arithmetic argument treats them.

1 Statement and exact premises

All variable counts include the input. Heights of integer polynomials are maximum absolute coefficients after coefficient parameters have been substituted. Put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad D = d^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = (a, b) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

The retained arithmetic premises have the following scope.

Elliptic group. We assume $2E(K) = \mathbb{Z}P$ and P non-torsion. Hence every point has an index in $\frac{1}{2}\mathbb{Z}$, as proved in [1, Corollary 1.3]. The reciprocal pair below is not restricted to $2E(K)$.

Pole obstruction and realization. For $b_0 \neq 0$ let $B_0 = 17b_0$. Rational solutions of

$$p_0^2 = B_0(1 - e_0)(1 + 3e_0), \qquad B_0 q_0^2 = (s_0^2 - B_0 e_0(B_0 - b_0^2))(s_0^2 + 3B_0 e_0(B_0 - b_0^2)) \tag{1}$$

exclude negative odd valuations of b_0 outside $2, 3, 17$. We retain the converse with $p_0 q_0 s_0 \neq 0$ for prepared inputs

$$\begin{gathered} 0 < b_0 < 17, \qquad b_0 \text{ has square reduced denominator,} \\ b_0 \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b_0) \geq 3, \quad v_3(b_0) \geq 1, \quad v_{17}(b_0) \geq 2. \end{gathered} \tag{2}$$

These are the same obstruction and prepared nonzero-realization premises as the predecessor [3]; their rank and descent foundations are not independently re-proved here.

Prime selection. We retain the fixed-data simultaneous-prime statement of [5, 6]. It applies to any fixed finite list of distinct rational contact coefficients after a fixed square clearing of denominators. It allows prescribed divisibility of the square denominator, finite avoidance, an arbitrarily large fixed lower bound on the rational argument, and the Gaussian congruences and conjugate annihilating orders needed for the common index. In the source normalization these orders U_*, V_* and preparation modulus L_* have

$$\gcd(U_*, V_*) = 1, \quad \gcd(U_*, L_*) = 16, \quad \gcd(V_*, L_*) = 1,$$

so $h \equiv 0 \pmod{U_*}$, $h \equiv 1 \pmod{V_*}$ and $h \equiv h_* \pmod{L_*}$ are compatible when $64 \mid h_*$. The source also retains the prescribed compact real neighborhoods in that progression. All finite data and rational coefficients are fixed before prime selection. We check the new coefficient list and all subsequent uses in Section 5.

Additional hypothesis. Ordinary rational *abc* is assumed. The eight-contact estimate below is not asserted to follow from the separate weak-*abc*/Szpiro hypothesis of the thirteen-variable family. No modular contact-height premise is needed here.

Theorem 1.1. *Under these premises there is a computable family* $P_B \in \mathbb{Z}[X, Y_1, \dots, Y_{80}]$, *indexed by odd integers* $B \geq 3$, *such that every member has degree four, content one, 941 nonzero monomials, and height* $\max(4153456, B^2 + 797551)$. *There is a fixed* B_* *such that for every odd* $B \geq B_*$,

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{80} : P_B(x, \boldsymbol{y}) = 0.$$

The coefficient parameter B *is not a quantified variable. No numerical sufficient* B_* *is given; the* $B = 3$ *table is an algebraic fixture, not a certified defining member.*

2 The quadratic system

A field variable denotes two rational coordinates in the basis $1, \rho$. A bar denotes conjugation. The notation S^2 denotes an ordinary product, whereas the variables named F_H, G_H, F_U, F_V are stored field values. Every displayed equation has degree at most two in the retained rational coordinates.

2.1 The common reference point

Put $A_P = a^2 + D$, $B_P = a^2 - D$. Retain $S, T_H, f_H, g_H \in K$ and impose

$$2bD(aT_H - b)f_H = aA_PST_H + bA_PS - bB_PT_H + 4a^2D, \tag{3}$$

$$2bD(aS + b)g_H = -aA_PST_H - bB_PS + bA_PT_H - 4a^2D, \tag{4}$$

$$D^2 f_H g_H + aD(f_H + g_H) + aST_H + b(S - T_H) + D = 0. \tag{5}$$

These are the complete fixed-difference equations of [4]. They give $H, H - P$ finite with nonzero abscissas and

$$S = y(H)/x(H), \quad T_H = y(H - P)/x(H - P), \quad f_H = 1/x(H), \quad g_H = 1/x(H - P).$$

In particular $f_H g_H \neq 0$, but we do not assume $ST_H \neq 0$ on all components.

For explicitness, the exceptional-leading-coefficient check is as follows. Put

$$\Phi = aS^2T_H^2 - (a^2 - D)(S^2 + T_H^2) + 2(a^2 + D)ST_H + 4aD,$$

and define

$$2bDf_0 = -(a^2+D)S + (a^2-D)T_H - bS^2 - aS^2T_H,$$
$$2bDg_0 = (a^2+D)T_H - (a^2-D)S - bT_H^2 + aST_H^2.$$

With L_f, L_g the two leading linear coefficients, and r_f, r_g the first two residuals, the last residual R satisfies

$$L_fL_gR = D^2r_fr_g + (aDL_g + D^2Q_g)r_f + (aDL_f + D^2Q_f)r_g - aD^2C_P\Phi,$$

where Q_f, Q_g are the displayed right sides and $C_P = D^2 + 6a^2D - 3a^4 \neq 0$ (its norm is $-8203357319351/16777216$). Hence $\Phi = 0$ at every solution. If $L_fL_g \neq 0$, then $f_H = f_0, g_H = g_0$. If $L_f = 0$, then $T_H = b/a$, $L_g = -aDC_P/(a^2+D) \neq 0$, $g_H = 1/a$, and the last equation has nonzero f_H coefficient $D(a^2+D)/a$. It again gives $f_H = f_0$. The other case is symmetric. The identities

$$Df_0^2 + S^2f_0 - 1 = \frac{a^2S^2 - b^2}{4ab^2D}\Phi, \qquad Dg_0^2 + T_H^2g_0 - 1 = \frac{a^2T_H^2 - b^2}{4ab^2D}\Phi$$

exclude zero reciprocals. The chord identity $a(f_0T_H + g_0S) + b(g_0 - f_0) = S + T_H$, together with the abscissa chord formula, gives $H - P$; when $x(H) = a$, the only allowed case is $H = -P$, and the tangent formula gives $H - P = -2P$. The converse follows from the same identities. Thus no open-subset assumption is hidden in (5).

2.2 Two points and their difference

Retain $f, h, g, k, s \in K$, with stored squares F_U, F_V, and impose

$$F_U = f^2, \quad F_V = g^2, \quad h^2 = f - DfF_U, \quad s(f-g) = h+k, \quad s(h-k) = 1 - D(F_U + fg + F_V). \tag{6}$$

The complete algebraic proof in [1, Proposition 2.1] identifies points $U, V \in E(K)$, including the identity, with $U \neq V$. Their finite difference $A = U - V$ has

$$c = x(A) = s^2 + D(f+g), \qquad y(A) = Dh - s(Df + c).$$

Neither U, V nor A is assumed to have square abscissa. The theorem includes the tangent case $f = g$, and excludes $(f, h) = (g, k) = (0, 0)$. Define one retained rational coordinate

$$T = 2\,\mathrm{Re}(s^2 + D(f+g)) = \mathrm{Tr}\, x(A). \tag{7}$$

This is a quadratic equation. The index n of A belongs to $\frac{1}{2}\mathbb{Z}$, and may be zero.

2.3 Ratios, comparisons and the additive scale

Retain $L, L_0 \in K$ with

$$hL + fS = 0, \qquad (1 - DF_U)L + hS = 0, \tag{8}$$
$$kL_0 + gT_H = 0, \qquad (1 - DF_V)L_0 + kT_H = 0. \tag{9}$$

At formal nonidentity points these are the usual ratios, and at an identity numerator they force the ratio to be zero. No division by h or k is made on torsion components.

Retain the input $X \in \mathbb{Q}$, a comparison coordinate $I_0 \in \mathbb{Q}$ and a slope $\lambda \in \mathbb{Q}$. Write $L = L_r + \rho L_i$ and impose

$$I_0 - L_r + X^2 + B = 0, \qquad \beta = L_r - I_0 = X^2 + B, \qquad I = I_0 + \rho L_i, \qquad C = L - \overline{L_0}.$$

Retain $F_H, G_H, C_2, I_2 \in K$ and two scalar height coordinates u_h, v_h, with

$$F_H = f_H^2, \quad G_H = g_H^2, \quad C_2 = C^2, \quad I_2 = I^2, \qquad u_h = \beta^2 + T^2, \quad v_h = \beta T.$$

The scale $a_1 = \beta^4 + T^4 = u_h^2 - 2v_h^2$ is not retained as a coordinate. Use arguments

$$t_0 = \lambda, \qquad t_1 - \lambda = u_h^2 - 2v_h^2. \tag{10}$$

Only t_1 is a further coordinate. Notice that $a_1 \geq B^4 \geq 81$.

2.4 Contact and pole equations

For each argument $t = t_j$ retain scalars R, Z, w, w_2, w_3, field values $A_1, A_2, \Omega = \omega_0 + \rho\omega_1$, and scalars $\eta, b_0, e, u, p, v, q, p_2, q_2$. Impose

$$R = t^2 - 2t, \qquad RZ + Z - R^2 + 35R = 0, \qquad (R+1)w = Z^2 + 24Z, \tag{11}$$

$$w_2 = w^2, \qquad w_3 = ww_2, \tag{12}$$

$$w_3 A_1 = w_2^2 + w_3 C_2 + w_2 F_H + F_U^2, \tag{13}$$

$$w_3 A_2 = w_2^2 + w_3 I_2 + w_2 \overline{G_H} + \overline{F_V}^{\,2}, \tag{14}$$

$$(R+1)\Omega = A_1 A_2, \tag{15}$$

$$\omega_0^2 - 2\omega_0\eta - 2\omega_1^2 = 0, \qquad w_3 b_0 = \eta^2, \tag{16}$$

$$p_2 = p^2, \qquad 17 b_0 p_2 = 289 + 34e - 3e^2, \tag{17}$$

$$e(17 - b_0) = 17uv, \qquad q_2 = q^2, \tag{18}$$

$$17 b_0 q_2 = u^2 + 34uv - 867v^2. \tag{19}$$

In each factor equation a scalar is interpreted as a field element. The two blocks have separate pole coordinates. For $b_0 \neq 0$ the last five equations imply (1) with

$$e_0 = e/17, \quad p_0 = b_0 p, \quad s_0 = b_0 u, \quad q_0 = b_0^2 uq.$$

Conversely a prepared witness with $s_0 \neq 0$ gives these coordinates, using $v = e(17 - b_0)/(17u)$. Zero values on other components are not silently divided out.

3 Domains and local extraction

Put $r = R + 1$, $z = Z + 37$. The contact equations give

$$r = (t-1)^2, \quad r(z - r) = 36, \quad rw = z^2 - 50z + 481.$$

Thus $r \neq 0$. With

$$P_4(r) = (r-1)(r-4)(r-9)(r-36), \qquad \Theta = \{-5, -2, -1, 0, 2, 3, 4, 7\},$$

one has $P_4(r)/r^2 = z^2 - 50z + 481$ and $F(t) := \prod_{\theta \in \Theta}(t - \theta) = wr^3$. If $w = 0$, the factor equations give $F_U^2 = \overline{F_V}^{\,2} = 0$, hence $f = g = 0$; the cubic gives $h = 0$, the secant gives $k = 0$, and the last equation of (6) gives $0 = 1$. Therefore $rwF(t) \neq 0$ at every rational solution, and $t \notin \Theta$.

For interpretation only, define

$$q_* = w^{-1} = \frac{(t-1)^6}{F(t)}, \quad R_* = \frac{F(t)}{(t-1)^{10}}, \quad T_i = A_i/w, \quad W = T_1 T_2 = \frac{r}{w^2}\Omega, \quad \tau = \frac{r}{w^2}\eta.$$

Then

$$T_1 = 1 + q_* C^2 + q_*^2 f_H^2 + q_*^4 f^4,$$
$$T_2 = 1 + q_* I^2 + q_*^2 \overline{g_H}^{\,2} + q_*^4 \overline{g}^{\,4}, \qquad b_0 = R_* \tau^2.$$

The cleared trace equation is preserved under this common rescaling. None of W, η, b_0 is assumed globally nonzero at this stage.

For $t \notin \Theta$, let $M(t)$ be the product of primes $p > 76176$ with positive odd valuation at one of $t - \theta$. At such a prime, all contact differences and $t - 1$ are units. Write $\nu = v_p(F(t)) > 0$, odd; then $v_p(q_*) = -\nu$. Every nonzero reciprocal abscissa has even valuation at these good places. This follows directly from $h^2 = f - Df^3$ if $v(f) \neq 0$; valuation zero is even too. The same statement holds for the common reference point. A zero reciprocal contributes no nonconstant term.

The finite nonconstant valuations in each factor are respectively odd, 2 modulo 4, and 4 modulo 8, and differ from the constant term of valuation zero. Thus both factors are nonzero and have valuation at most zero at every place above p. Let $\ell = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \leq 0$. The trace equation forces the rational component of W and τ to be nonzero, and

$$\tau = \operatorname{Tr}(W^{-1})^{-1}, \qquad v_p(b_0) = \nu - 2v_p \operatorname{Tr}(W^{-1}) \leq \nu + 2\ell.$$

This valuation is odd. The pole obstruction forces it positive, so $\ell > -\nu/2$. At a place attaining ℓ, each nonconstant summand valuation exceeds $-\nu/2$. Consequently

$$v(C), v(I) > \nu/4, \quad v(f_H), v(\overline{g_H}) > 3\nu/4, \quad v(f), v(\overline{g}) > 7\nu/8.$$

The reciprocal valuations are positive even integers, and C, I have order at least one. In particular the four indicated points are in the appropriate primary or conjugate formal groups, with the identity convention at zero reciprocals.

Write the indices of H, U, V, A as $h_*, u_*, u_* - n, n$ in $\frac{1}{2}\mathbb{Z}$. At a consumed prime, $H, H - P$ are nonidentity formal points and therefore non-torsion. Multiplying the integer-index group identities by two kills every possible two-torsion component. The full formal-logarithm ratios are

$$u_*/h_*, \qquad (u_* - n)/(h_* - 1).$$

At least one denominator is a p-adic unit, and the numerator indices are p-integral. Thus one full ratio is integral. The order-four relative logarithm estimate in [1] first makes its plain ratio integral. The test $C \equiv 0 \pmod{p}$ then gives integrality of the other plain ratio, hence of its full ratio. Only after this step are the relative errors used as absolute congruences. Subtracting gives

$$n = h_* \frac{u_*}{h_*} - (h_* - 1)\frac{u_* - n}{h_* - 1} \equiv L \equiv \beta \pmod{p}.$$

This proves that, if $\beta \neq n$ and N is the positive reduced numerator of $\beta - n$, then

$$M(t_0) \mid N, \qquad M(t_1) \mid N. \tag{20}$$

Inert primes are included: if the required formal groups are incompatible, no solution occurs at that prime. No assertion that the two radicals are coprime is needed.

4 Heights and the uniform sufficient shift

Lemma 4.1. *Under rational abc, fixed constants C_0, C_1 satisfy*

$$h(t) \leq \tfrac{1}{3} \log M(t) + C_0 \quad (t \notin \Theta), \qquad h(1 : u : v) \leq \tfrac{4}{3} h(u^4 + v^4) + C_1 \quad (u, v \in \mathbb{Q}).$$

Proof. For $t = a/b$ reduced use $G(a,b) = \prod_{\theta\in\Theta}(a-\theta b)$. Granville's Proposition 1 [2], with exponent loss 1/2, gives $\operatorname{rad}(G) \gg \max(|a|,b)^{11/2}$. Since $\operatorname{rad}(m)^2 \le |m|\operatorname{sf}(m)$ and $|G| \ll \max(|a|,b)^8$, its odd part is at least a fixed multiple of $\max(|a|,b)^3$. Outside the fixed exceptional primes it is exactly $M(t)$: two primitive contact factors cannot vanish together, and a prime in b divides none. This proves the first assertion, including bounded heights by increasing C_0. The form has eight distinct factors and $G(a,b) \neq 0$, so all source hypotheses are met. The author's corrigendum changes Theorems 8.1 and 8.3, not this proposition; neither corrected theorem is used here.

For the second assertion, the zero-coordinate cases are immediate. Otherwise write $u = gA/D_0$, $v = gB_0/D_0$ with $\gcd(A,B_0) = \gcd(g,D_0) = 1$, $g, D_0 > 0$. Set $H = \max(|A|,|B_0|)$, $N_0 = A^4 + B_0^4$, and $k = \gcd(N_0, D_0^4)$, $m_0 = N_0/k$, $n_0 = D_0^4/k$. Then $u^4 + v^4 = g^4 m_0/n_0$ is reduced. Put $M = \max(g^4 m_0, n_0)$. Rational *abc* at exponent 16/15 and $\operatorname{rad}(N_0) \le D_0 m_0$ give $H^{7/4} \le K^{15/16} D_0 m_0$. Also $D_0^4 \le 2n_0 H^4/m_0$, whence $H^{3/4} \le 2^{1/4} K^{15/16} m_0^{3/4} n_0^{1/4}$. Finally $\max(gH, D_0) \le 2^{1/4} H (M/m_0)^{1/4}$. Taking logarithms bounds $h(1:u:v)$ by a fixed constant plus $\frac{3}{4}\log m_0 + \frac{1}{3}\log n_0 + \frac{1}{4}\log M \le \frac{4}{3}\log M$. This accounts for all common numerator and denominator cancellation. □

Suppose $\beta \neq n$. By (10), (20) and the elementary additive height inequality,

$$h(a_1) \le h(t_0) + h(t_1) + \log 2 \le \tfrac{2}{3}\log N + 2C_0 + \log 2.$$

Writing $H_2 = h(1:\beta:T)$, the preceding lemma yields

$$H_2 \le \tfrac{8}{9}\log N + C_2.$$

Because $2n \in \mathbb{Z}$, the reduced numerator satisfies $N \le 2(1+|n|)e^{h(\beta)}$. Hence

$$H_2 \le 8\log(2(1+|n|)) + 9C_2.$$

The finite point $A = U - V$ satisfies the all-point trace theorem [1, Corollary 1.3], including torsion and half-integral indices. Thus

$$n^2 \le 8A_0\log(2(1+|n|)) + 9A_0C_2 + J_0.$$

This bounds $|n|$ by a fixed constant and then bounds H_2 by a fixed constant, independent of B and every witness. But $h(\beta) \ge \log\beta \ge \log B$, since $\beta = X^2 + B$. One sufficiently large fixed B_* excludes $\beta \neq n$ for all $B \ge B_*$. Therefore $\beta \in \frac{1}{2}\mathbb{Z}$. Writing $X = a/b$ reduced gives $b^2 \mid 2a^2$, hence $b^2 \mid 2$ and $b = 1$. This proves soundness. The replacement of $2X^2 + B$ by $X^2 + B$ is essential for this denominator argument with half-indices.

5 Completeness

Fix an integer X and an odd $B \ge 3$. Set $n = \beta = X^2 + B > 0$ and $A = [n]P$. This fixes $T = \operatorname{Tr} x(A)$ and $a_1 = \beta^4 + T^4 \ge 81$ before selecting primes. Choose a nonexceptional reference multiple $H = [h]P$, with its initial index divisible by 64, in fixed compact real and finite-place neighborhoods. Set $U = [hn]P$ and $V = [(h-1)n]P$. All are in the doubled group, so the usual square-abscissa assertions are available for these particular witnesses, although they were not imposed on all solutions. Avoid finitely many forbidden reference points and denominators. The reciprocal-pair secant or tangent lemma then supplies its coordinates; A is never the identity.

The new fixed list of contact coefficients in the slope variable is

$$\Theta \;\cup\; (\Theta - a_1).$$

The two lists are disjoint since their span is 12 and $a_1 \geq 81$. Choose an integer $D_1 > 0$ whose square clears these rational coefficients, sufficiently divisible at every preparation prime. Apply the retained prime-selection input to this list with $\lambda = u/(D_1^2 w^2)$. Its prime numerators are exactly

$$u - \theta D_1^2 w^2, \qquad u + (a_1 - \theta) D_1^2 w^2 \qquad (\theta \in \Theta).$$

The zero contact in the first list makes u one selected prime. The finite avoidance data include all denominators and contact collisions. The prescribed conjugate reduction orders and the compatible common-index progression make H, U formal at the primary places, and $H - P, V$ formal at the opposite places. Both full ratios equal n, so the reciprocal tests have depth at least two and C, I have depth at least four. With $v(q_*) = -1$, every nonconstant factor term is positive at the selected place. At the opposite place each factor is either a unit of residue one or has negative valuation. Thus $\mathrm{Tr}(W^{-1})$ reduces to one or two and is a unit. The actual pole input has valuation one there.

Put $v_0 = D_1^2 w^2$. For $a \in \{0, a_1\}$ the exact identity is

$$R_*(\lambda + a) = \frac{\prod_{\theta \in \Theta}(u + (a - \theta)v_0)}{v_0^8 (\lambda + a - 1)^{10}}.$$

All factors other than the selected prime numerators are rational squares. The trace changes every valuation by an even number. Consequently the pole input has square reduced denominator. At $2, 3, 17$ choose denominator divisibility so the argument dominates the fixed a_1 p-adically. Then $q_*, R_* \sim t^{-2}$, all weighted tests are small, and $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two. This gives the required local squareclasses and depths in (2). On the compact real neighborhoods, choosing an arbitrarily large fixed lower bound on λ makes both arguments large and positive; $W \to 1$ and $b_0 \sim 1/(4t^2)$, so $0 < b_0 < 17$. These bounds are chosen before the prime selection. Independent prepared pole witnesses now fill both blocks.

No parity restriction on n was used: multiplication by an even or odd integer preserves the formal group, and the full ratios are the same integer n. The construction works for $X = 0$ as well. It proves completeness for every integer input and every allowed fixed shift. Together with the uniform soundness threshold it proves Theorem 1.1.

6 Counts, coefficients and verification

The retained rational-coordinate census is

Input, real comparison and slope	3
Common reference chart	8
Reciprocal pair and secant parameter	10
Two field ratios	4
Four reciprocal squares and two comparison squares	12
Trace and two height outputs	3
Two contact/pole blocks	40
Second argument	1
Total	81

There are 73 rational quadratic equations. Clear each to a primitive integer row with positive first coefficient in the specified order. For the first six common-chart rows apply the fixed invertible rational matrix in `common-row-reduction.json`, whose determinant is $-1/362963666515968$, and normalize again. This changes no zero set. The sum of the 73 row squares is P_B.

The directory `computation/quartic/abc81_v25` contains the complete coordinate and row table and its fully expanded quartic. Exact collection gives 942 parameter-separated terms and 941 coordinate monomials for every positive B. The coefficient of X^4 is one, proving exact degree and content. The constant coefficient is $B^2 + 797551$; the largest fixed coefficient is 4153456. Every remaining parameter coefficient is a linear multiple of B, bounded by the maximum of these two values. Thus the stated height formula is exact. For odd $B \geq 1833$ it is $B^2 + 797551$; this is only a coefficient threshold.

A separate implementation reconstructs all 73 equations directly from the field formulas using symbolic polynomial arithmetic, checks the rational row matrix, and independently expands every quartic coefficient. It checks the universal point and contact identities and rejects a changed coefficient, missing row and corrupted quartic. The 702 exact elliptic examples described in [1] include the exceptional branches used here. The computations do not establish *abc*, the inherited arithmetic premises, a numerical defining shift, or non-author acceptance of the proof.

References

[1] Companion paper, *Trace heights on quadratic twists with $j = 1728$*, 13 November 2026, `papers/trace_heights/trace_on_all_points_20261113_v25.tex`.

[2] Andrew Granville, *Rational and integral points on quadratic twists of a given hyperelliptic curve*, International Mathematics Research Notices (2007), rnm027, Proposition 1; author's version, printed p. 6, https://dms.umontreal.ca/~andrew/PDF/hyperelliptics.pdf. See also https://dms.umontreal.ca/~andrew/PDF/hyperellCorr.pdf; its corrections to Theorems 8.1 and 8.3 are not used here.

[3] Companion paper, *An 87-variable quartic under abc*, `papers/quartic_existence/quartic_abc87_20261110_v21.tex`, repository commit `fa09c9c6c97982c13265e7088d95cdb61b3e90cf`.

[4] Companion paper, *Fixed-difference elliptic charts*, `papers/elliptic_charts/fixed_difference_charts_20261110_v18.tex`.

[5] Companion paper, `papers/arithmetic_dependencies/arithmetic_dependencies_20261108_v8.tex`.

[6] Companion paper, `papers/prime_selection/prime_selection_20261109_v12.tex`.